

Results: Thermal Conductivity Measurements Across Sample Batches

Materials Characterization Group

1 Introduction

This document reports thermal conductivity, density, and tensile strength measurements collected across four sample batches of a composite laminate material. All measurements were performed under controlled laboratory conditions, with uncertainties reported at the one-standard-deviation level unless otherwise noted. Section 2 reports the conductivity results using the modern `siunitx` unit macros, while Section 3 reports density and strength results using the legacy `\SI` and `\si` macros for comparison, since both forms remain common in the literature this study draws on.

2 Thermal Conductivity

Table 1 summarizes the measured thermal conductivity for each batch at three test temperatures, given as 25 °C, 75 °C, and 125 °C. Values are reported in $\text{W m}^{-1} \text{K}^{-1}$, with sample counts and standard deviations given alongside each mean.

Table 1: Thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$) by batch and temperature.

Batch	25 °C	75 °C	125 °C
A1	0.842	0.901	0.958
A2	0.831	0.889	0.947
B1	0.917	0.972	1.034
B2	0.905	0.963	1.021

Across all four batches, conductivity increases approximately linearly with temperature over the tested range, and batches B1 and B2 consistently outperform A1 and A2, which is consistent with the higher filler fraction used in the B-series formulation. The largest single measurement, $1.034 \text{ W m}^{-1} \text{K}^{-1}$ for batch B1 at 125 °C, remains well within the manufacturer’s rated operating envelope of $1.2 \text{ W m}^{-1} \text{K}^{-1}$.

Table 2 reports summary statistics across all batches combined, including the sample count, mean, and standard deviation at each temperature, using `multicolumn` and `cmidrule` to group the per-temperature statistics under shared headers.

The standard deviation at each temperature is small relative to the mean, of order 4×10^{-2} , indicating good repeatability across the four batches. We note that the sample size per batch was $n = 12$ individual coupons, so the batch means reported in Table 1 are themselves averages over a reasonably large number of measurements, and the pooled standard deviations in Table 2 should be interpreted accordingly.

Table 2: Summary statistics across all batches.

	25 °C			75 °C			125 °C		
	<i>n</i>	Mean	SD	<i>n</i>	Mean	SD	<i>n</i>	Mean	SD
All batches	4	0.874	0.041	4	0.931	0.040	4	0.990	0.041

3 Density and Tensile Strength

For comparison with legacy laboratory notebooks, this section reports density and tensile strength using the older and macro forms. Density was measured as 1.58 g cm^{-3} for the A-series batches and 1.64 g cm^{-3} for the B-series batches, consistent with the higher filler loading noted above. Table 3 reports tensile strength alongside elongation at break for all four batches.

Table 3: Tensile strength and elongation at break.

Batch	Strength (MPa)	Elongation (%)
A1	142.3	2.11
A2	138.7	2.04
B1	156.9	1.87
B2	151.2	1.92

The B-series batches show higher tensile strength but lower elongation at break than the A-series batches, a trade-off that is typical of increased filler loading in composite laminates: the additional filler stiffens the matrix and raises peak strength, at the cost of the ductility that allows the A-series material to elongate further before fracture. Combined with the conductivity results in Section 2, these measurements suggest that batch selection should be guided by the relative importance of thermal performance versus mechanical ductility in the intended application, since no single batch dominates on every axis studied here.

4 Conclusion

Across all measured properties, the B-series formulation offers higher thermal conductivity and tensile strength than the A-series formulation, at the cost of reduced elongation at break. These results, summarized in Tables 1, 2, and 3, will inform the batch selection guidance issued alongside the next revision of the material datasheet.